Krithi Iyer

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EDUCATION

Georgia Institute of Technology, Atlanta, GA | Anticipated Graduation: 2028 | GPA: 4.0/4.0
***Bachelor of Science* in Electrical Engineering,** *Threads: Bioengineering & Signal Processing*
Coursework: Linear Algebra, Data Structures & Algorithms, Circuit Analysis, Digital System Design, Intro Signal Processing

PROJECTS

Force-Sensitive Resistor Calibration Circuit, *Medical Robotics Club* August 2024 – Present

- ***Force Sensor Team Lead:*** Led development of a multi-sensor tactile sensing array for a low-cost prosthetic hand, enabling force measurement across individual fingers to approximate proprioceptive feedback during grasping tasks.
- Architected and prototyped an FSR-based sensing circuit using an Arduino Nano, integrating and calibrating 5+ force sensors (0.3 – 10 N range) to generate reliable analog force signals.
- Developed embedded firmware to acquire and process sensor signals, implementing ADC sampling and Python-based calibration curve generation to map raw analog readings to force estimates for downstream control analysis.

Predictive IV Drip Monitoring System, *Medtech Hackathon (2nd place)* November 2025 – February 2026

- ***Circuit Design Lead:*** Designed a sensor-driven system to detect and predict IV bag depletion and failure modes (occlusion, leak), enabling early alerts to reduce treatment interruptions in clinical settings.
- Developed a multi-sensor sensing architecture integrating fluid-level and optical sensors with an Arduino Uno for continuous monitoring of IV drip conditions.

Neuronal Modeling & Analysis, *Independent Computational Project* December 2025 – Present

- Simulated leaky integrate-and-fire neurons with spike-rate adaptation in Python (NumPy, Matplotlib), using Euler integration (1ms timestep) to analyze spiking responses to sinusoidal input stimuli.
- Implemented spike-triggered averaging and biologically inspired synaptic learning rules to explore how neurons develop feature selectivity from input spike patterns.

Radiation-Resistant Wound Dressing, *NASA L'SPACE NPWEE* May 2025 – August 2025

- ***Materials Scientist:*** Co-led materials research and proposal development for radiation-resistant wound dressing for deep space, synthesizing findings with subject-matter expert to refine technical rationale.

RESEARCH EXPERIENCE

Singer Lab, *Georgia Institute of Technology* January 2025 – Present

- ***Imaris Analysis Undergraduate Lead:*** Quantified microglia, astrocyte, and amyloid-β plaque colocalization from 40x confocal image stacks in Imaris under chronic unpredictable stress.
- Optimized deep-learning filament-generation models for microglial morphology by refining training datasets and correcting AI-generated segmentation errors to improve filament accuracy.
- Authored 3 analysis protocols and trained 2 undergraduate researchers to enhance workflow reproducibility.

Comparative Neuromechanics Lab, *Georgia Institute of Technology* November 2024 – Present

- ***Undergraduate Student Researcher:*** Corrected DeepLabCut rat gait data (17 landmarks, 10,000+ frames) in XMALab to ensure kinematic modeling accuracy.

Moorjani Lab, *University of Washington* June 2023 – June 2024

- ***Research Intern:*** Designed a high-resolution scoring system to quantify motor recovery after cervical spinal cord injury, validated data from 80+ rats and demonstrated enhanced sensitivity of the scoring system over conventional motor assessment scales.

PUBLICATIONS & PRESENTATIONS

Rahmati, S., Plemmons, M., Watermeier R., Iyer K., *et al.* "Automatic Registration of Generic Bone Models for X-Ray—Based Biomechanics: Application to 3D Rat Hindlimb Kinematics and Sex-Specific Gait Patterns." *Submitted to Computers in Biology and Medicine, 2025.*
Iyer, K., *et al.* "Mitigating Stress-Driven Potentiation of Amyloid Pathology via 40 Hz Sensory Flicker Stimulation." *Fall Undergraduate Neuroscience Research Symposium, Georgia Institute of Technology, 2025.*
Iyer, K., Moorjani, S. "A High-Resolution Behavioral Scoring System for Quantifying Motor Function after Spinal Cord Injury." *Manuscript in preparation.*

TECHNICAL SKILLS

Programming & Analysis: Python (NumPy, Pandas, Matplotlib, scikit-learn, Streamlit), Java, C++ (Arduino), MATLAB, Jupyter
Computer Vision & Imaging: DeepLabCut, XMALab, Imaris (3D segmentation, surface rendering, colocalization analysis)
Hardware & Design: Fusion 360, SolidWorks, NX, Arduino microcontrollers, analog circuit design, soldering, 3D printing, Onshape